James Madison University

Information Technology (IT) Program

IT 215 Intro to Telecom, Networking & Security

Fall 2025

Lab 3 - Dynamic Host Configuration Protocol

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Honor Pledge: I have neither given nor received help on this lab that violates the spirit of the JMU Honor Code.

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1 Introduction

This lab focuses mainly on developing practical skills needed for virtual machines and Wireshark.

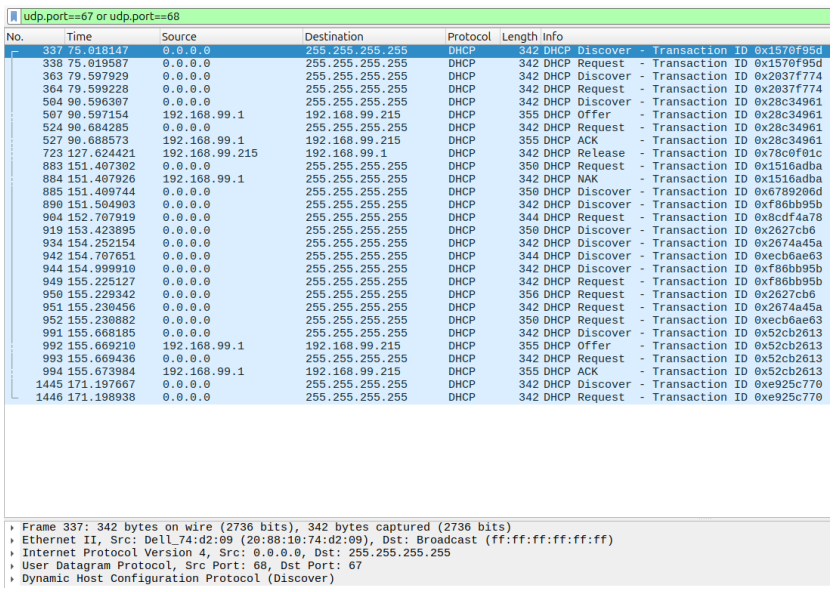
2 Lab Exercises: Network Access

2.1 Part A - Network Setup

T1: Step 1: Capture a Trace

Q1: Launch Wireshark and start a capture with a filter of "(udp port 67) or (udp port 68)". There is no shorthand to indicate DHCP, so we filter traffic using the UDP ports reserved for DHCP.(Provide a Snapshot)

A1: See Figure 1.



udp.port==67 or udp.port==68

No.	Time	Source	Destination	Protocol	Length	Info
337	75.018147	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x1570f95d
338	75.019587	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x1570f95d
363	79.597929	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x2037f774
364	79.599228	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x2037f774
504	90.596307	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x28c34961
507	90.597154	192.168.99.1	192.168.99.215	DHCP	355	DHCP Offer - Transaction ID 0x28c34961
524	90.684285	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x28c34961
527	90.688573	192.168.99.1	192.168.99.215	DHCP	355	DHCP ACK - Transaction ID 0x28c34961
723	127.624421	192.168.99.215	192.168.99.1	DHCP	342	DHCP Release - Transaction ID 0x78c0f01c
883	151.407302	0.0.0.0	255.255.255.255	DHCP	350	DHCP Request - Transaction ID 0x1516adba
884	151.407926	192.168.99.1	255.255.255.255	DHCP	342	DHCP NAK - Transaction ID 0x1516adba
885	151.409744	0.0.0.0	255.255.255.255	DHCP	350	DHCP Discover - Transaction ID 0x6789206d
890	151.504903	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xf86bb95b
904	152.707919	0.0.0.0	255.255.255.255	DHCP	344	DHCP Request - Transaction ID 0x8cdf4a78
919	153.423895	0.0.0.0	255.255.255.255	DHCP	350	DHCP Discover - Transaction ID 0x2627cb6
934	154.252154	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x2674a45a
942	154.707651	0.0.0.0	255.255.255.255	DHCP	344	DHCP Discover - Transaction ID 0xecb6ae63
944	154.999910	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xf86bb95b
949	155.225127	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0xf86bb95b
950	155.229342	0.0.0.0	255.255.255.255	DHCP	356	DHCP Request - Transaction ID 0x2627cb6
951	155.230456	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x2674a45a
952	155.230882	0.0.0.0	255.255.255.255	DHCP	350	DHCP Request - Transaction ID 0xecb6ae63
991	155.668185	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x52cb2613
992	155.669210	192.168.99.1	192.168.99.215	DHCP	355	DHCP Offer - Transaction ID 0x52cb2613
993	155.669436	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x52cb2613
994	155.673984	192.168.99.1	192.168.99.215	DHCP	355	DHCP ACK - Transaction ID 0x52cb2613
1445	171.197667	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe925c770
1446	171.198938	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0xe925c770

```
▸ Frame 337: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits)
▸ Ethernet II, Src: Dell_74:d2:09 (20:88:10:74:d2:09), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▸ Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
▸ User Datagram Protocol, Src Port: 68, Dst Port: 67
▸ Dynamic Host Configuration Protocol (Discover)
```

Figure 1: Verifying Content

Q1: When the capture is started, release and renew your IP address with the command given below. This procedure may cause your computer to lose network connectivity temporarily, and depending on the operating system it may disrupt network connections. To minimize the disruption, close any programs that are using remote servers and enter the commands into a local window. (Provide a Snapshot).

A1: See Figure 2.

T2: Step 2: Inspect the Trace

Q1: In this step and the steps that follow, we will inspect only the short DHCP exchange described above. This is because the traffic

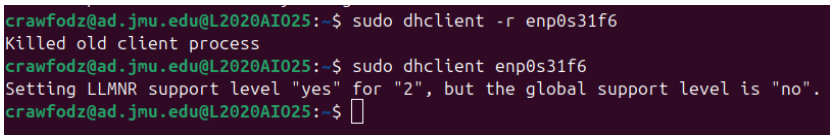


Figure 2: IP address

you have captured can vary widely across settings. You may have as few as two DHCP packets on a quiet network or many DHCP packets on a busy network (especially if a class is running this lab!). The details of DHCP packets may vary depending on how the computers implement DHCP. There may be multiple packets of a single kind in an exchange due to replicated servers, and different types of DHCP packets too. (Provide a Snapshot)

A1: See Figure 3.

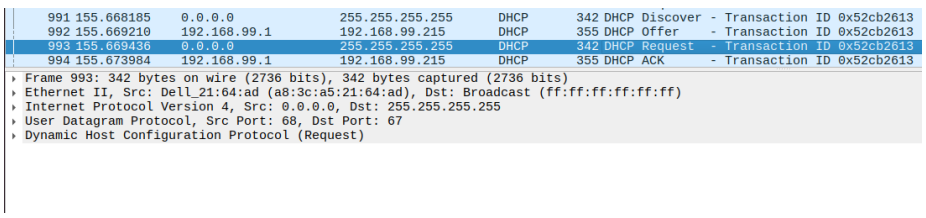


Figure 3: Python3 Installation

T3: Step 3: Details of DHCP Messages

Q1: What are the two values of the BOOTP Message Type field?

A1: 67 and 68 are the two values for the BOOTP message type.

Q1: How long is the Transaction ID field? Say whether it is likely that concurrent DHCP operations done by different computers will happen to pick the same Transaction ID.

A1: The transaction ID field is 0x52cb2613 and a length 6, which makes it unlikely that another computer will have the same transaction ID.

Q1: What is the name of the field that carries the IP address that is being assigned to the client? You will find this field filled in on the DHCP Ack, as that message is completing the assignment.

A1: The name of the field is "Requested IP Address" which in this situation is 192.168.99.215

Q1: What is the value of the Magic Cookie that stands for DHCP?

A1: The magic cookie value is DHCP, differentianting and sorting BOOTP messages from DHCP messages in network configuration.

Q1: The first DHCP option is DHCP Message Type. What option value stands for this type?

A1: The first option value is a request (3).

Q1: DHCP Requests will typically have a Client Identifier option. Look at the value of this option. How does it identify the client? Take a guess.

A1: There is no direct "Client Identifier" option, but the request does give information like client IP and client mac address. It lists this computer as the client (the device I pinged the access point from).

Q1: DHCP Acks will typically have a Server Identifier option. Look at the value of this option. How does it identify the server? Take a guess

A1: I assumed it used the IP address of the server to identify it, and I was correct. It has the server name and IP address present in the Server Identifier tab.

Q1: What option value stands for the Requested IP Address option? And for the IP Address Lease Time option?

A1: The value for Request IP Address is the local machine name and IP address. The lease time option value is 1 day (86400)

Q1: How does the recipient of a DHCP message know that it has reached the last option?

A1: It receives an option that has the end value (255)

T4: Step 4: DHCP Message Addressing

Q1: What port number does the DHCP client use, and what port number does the DHCP server use? Ports matter because UDP messages are addressed using ports. Both of these port numbers are on the Request in the source and destination port fields (and you will also see them on the Ack).

A1: The port number for the client is 67 and the port number for the server is 68.

Q1: What source IP address is put on the Request message? It is a special value meaning "this host on this network" used for initialization.

A1: The source IP address is 0.0.0.0

Q1: What destination IP address is put on the Request message? It is also a reserved value designed to reach the DHCP server wherever it is on the local network.

A1: Listed at "Next Server IP address is also put as 0.0.0.0. Clearly the system is using uniform values to establish a connection before using real information.

Q1: What source Ethernet address is put on the Request message, and what destination Ethernet address is put on the Request message? One of these addresses is a reserved address

A1: The destination Ethernet address is ff:ff:ff:ff:ff:ff, and the source Ethernet address is a8:3c:a5:21:64:ad. The destination address is clearly the reserved address.

Q1: How does a computer work out whether a DHCP message it receives is intended as a reply to its DHCP Request message, and not a reply to another computer? Hint: if you are not sure then go over the fields you inspected previously in Step 2 above.

A1: It will send a 255 message, which is an end request, telling the server not to reply to another computer. This is like ending the handshake.

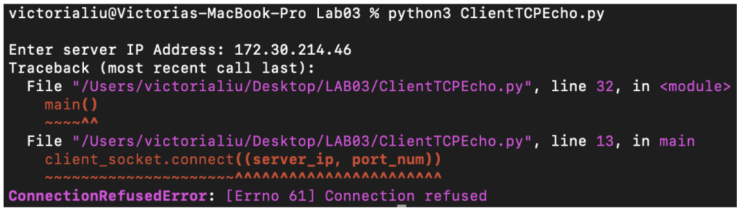
2.2 Part B - Socket Programming

Q1: Start your TCP client application without the TCP server running. What happens? Why?

A1: When trying to start my TCP client application without the TCP server running, it causes an "ConnectionRefusedError", this is because the client needs to connect to a listening server socket, because there was no listening server socket it immediately returned as an error showing that the port is closed and the connection was refused. See Figure 4.

Q1: Try different examples and explain the output!

A1: The terminal window on the top shows the server receiving all the messages sent from the client. The first input the client sent was "Hi Server", the server returned the message as "hI sERVER" this is due to the switching of cases. The client then sends "THIS IS FOR IT215", all the letters the server switched the uppercase to lowercase, however the numbers stayed the same as numerical digits are preserved. The third input "LaB 03" follows the same rules as the first two inputs. See Figure 5.



```
victorialiu@Victorias-MacBook-Pro Lab03 % python3 ClientTCPEcho.py

Enter server IP Address: 172.30.214.46
Traceback (most recent call last):
  File "/Users/victorialiu/Desktop/LAB03/ClientTCPEcho.py", line 32, in <module>
    main()
    ~~~~^^
  File "/Users/victorialiu/Desktop/LAB03/ClientTCPEcho.py", line 13, in main
    client_socket.connect((server_ip, port_num))
    ~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^
ConnectionRefusedError: [Errno 61] Connection refused
```

Figure 4: ConnectionRefusedError

Q1: Capture using WireShark the data transferred between Client and Server.

A1: The packet list shows the the TCP three way handshake SYN to SYN and ACK to ACK, establishing the connection and then the packet with PSH, ACK carries the client message. The server then sent a PSH, ACK packet containing the encoded response.See Figure 6.

Q1: Follow TCP Stream

A1: See Figure 7.

• Save the "Filtered" PCAP file. Submit the PCAP file with your report! • Submit both client and server code (script).

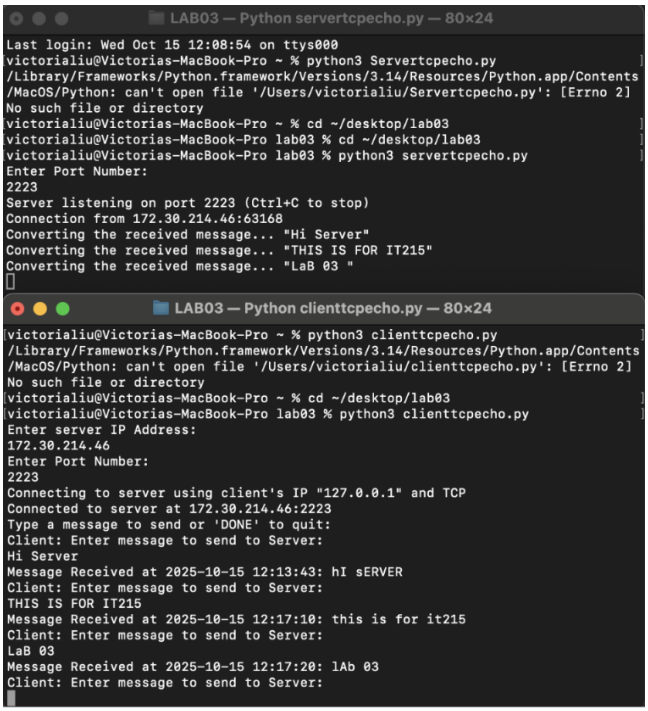


Figure 5: Client vs. Server Interaction

3 Conclusion

3.1 Summary

Overall, this lab strengthened my technical skills in Network Access and setup. We explored deeper into socket programming and how TCP works within client-server interactions.

3.2 Expectancy-Value-Cost

Expectancy
I felt pretty confident in completing the lab due to my prior knowledge of socket programming. It was refreshing to revisit socket programming and I felt like it was overall a straightforward lab with previous practice in WireShark as well.
Value
I think I gained valuable experience from this lab as I was able to connect more theoretical concepts about TCP and DHCP.
Cost
We put in around 7 hours of work on this lab.

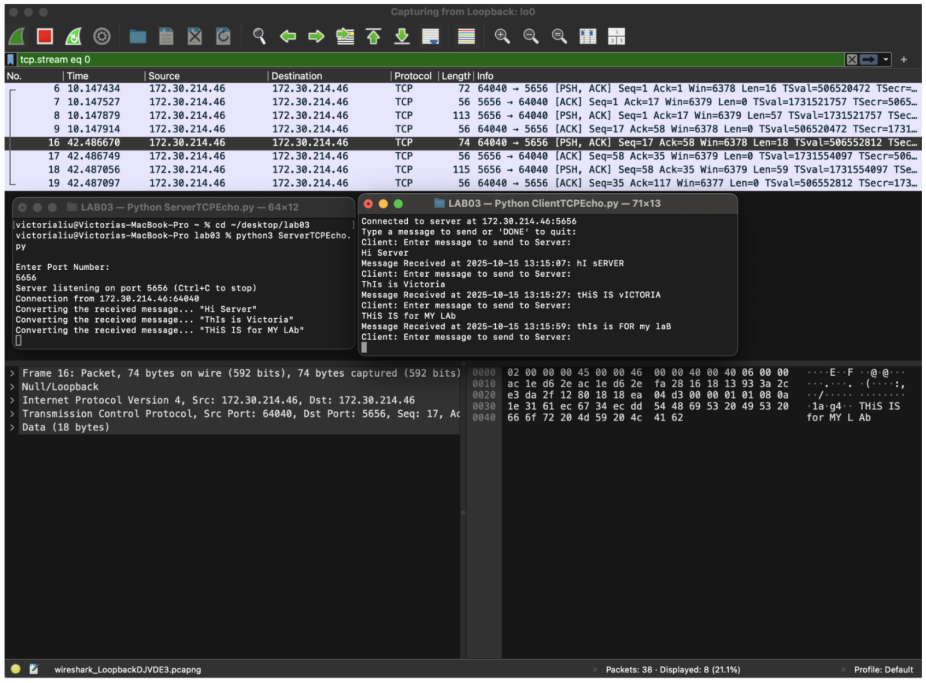


Figure 6: Captured Stream

3.3 Suggestions

Overall it was a straightforward lab, we completed it with ease.

3.4 Best Practices

Some best practices would be to read the questions carefully to fully understand what the problem is asking, and document step by step procedures.

4 Lab References

[1] OpenAI, *Response from ChatGPT: Wireshark Help.* Accessed: Oct. 15, 2025. Available: https://chat.openai.com/

[2] OpenAI, *Response from ChatGPT: Overleaf format help.* Accessed: Oct. 15, 2025. Available: https://chat.openai.com/

[3] OpenAI, *Response from ChatGPT: Socket Programming Assistance.* Accessed: Oct. 15, 2025. Available: https://chat.openai.com/

5 Acknowledgments

We would like to acknowledge Professor El-Tawab for providing us with TCP echo client and server examples.

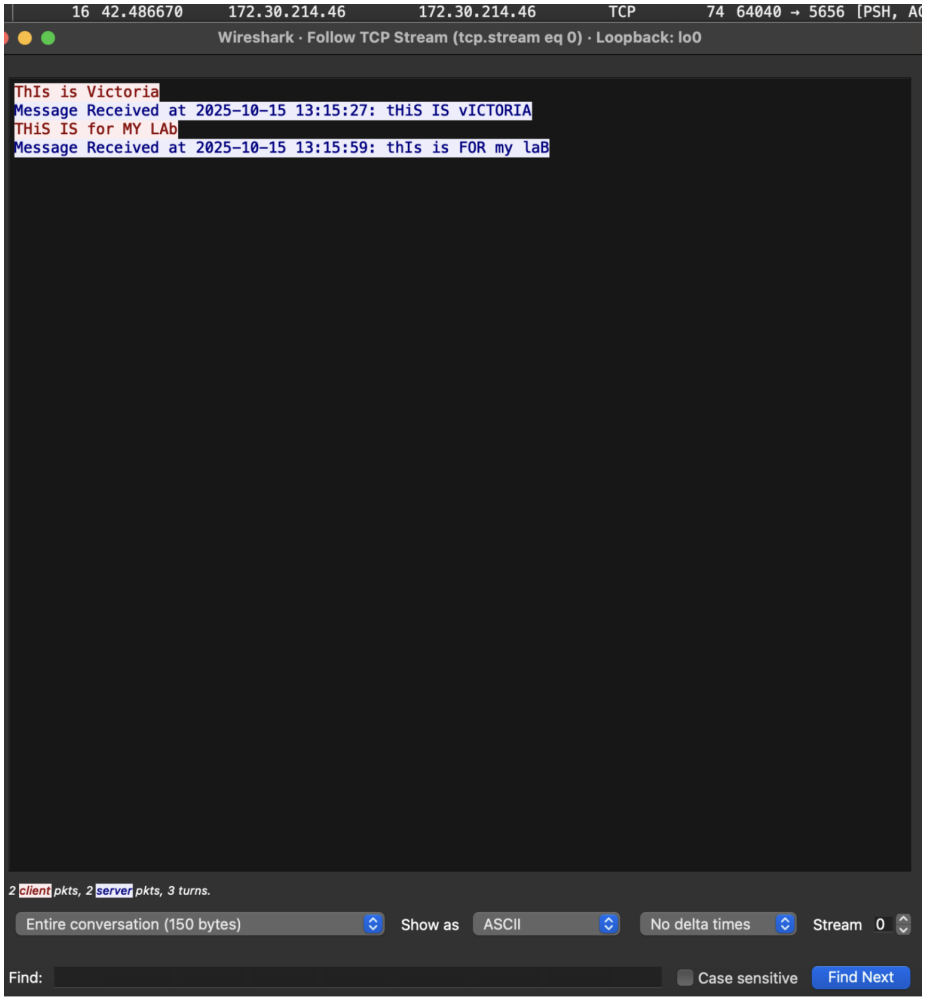


Figure 7: TCP Stream